

Q) Java Generates Hash Code for every object find out what is the hash code in Java & the purpose of Generating Hash Code.

Ans

Hash Code in Java : A hash code is a unique integer value generated by the hashCode() method for every object in Java. It is used to represent an object and helps in storing and searching objects efficiently.

Purpose of Generating Hash Code

- 1) It helps in fast searching & retrieval of objects
- 2) It is mainly used by Collection such as HashMap, HashSet, and Hashtable.
- 3) It improves the performance of data structures that use hashing
- 4) If two objects are equal (according to equals()), they should have the same hash code.

Ex

```
class Demo {
```

```
    public static void main (String [] args) {
```

```
        String s = "Java";
```

```
        System.out.println ("Hash Code:" + s.hashCode());
```

```
    }
```

```
}
```

O/P

Hash code : 2301506

NOTE :- The hash code is an integer value.

Different objects may sometimes have the same hash code (called a hash collision), but Java handles such cases internally.